SysMonitor

System Design Description

Abstract

This document describes the design and implementation of the SysMonitor support system of Arrowhead 5.2.2 (`ah5-SysMonitor-java-spring`).

Contents

1 Overview **3**

2 Implementation **4**
2.1 Implementation language and tools 4
2.2 Functional properties implementation 4
2.3 Non functional properties implementation 4
2.4 Deployment 5
2.5 Local modifications 5
2.6 Verification 5

3 Services **6**

4 References **6**

5 Revision History **8**
5.1 Amendments 8
5.2 Quality Assurance 8

1 Overview

This document describes the SysMonitor system implementation, which realizes the SysMonitor SysD [1]. SysMonitor collects the native `sysMonitor` observations of all systems from the AMQP broker, stores their history and republishes them on the MQTT broker (`sysMonitorData`). In Section 2, we describe implementation details of the system.

2 Implementation

This implementation realizes the SysMonitor SysD in `ah5-SysMonitor-java-spring`. The Java sources, dependencies, tests, image and deployment definition are owned by this folder. The previous implementation is preserved in `legacy/` and is excluded from the build.

2.1 Implementation language and tools

- Java 21 without a framework (JDK `HttpsServer`), Bouncy Castle, SQLite JDBC and RabbitMQ AMQP client; Eclipse Paho MQTT v5 output client. Maven produces only `arrowhead-sysmonitor-5.2.2.jar`; the fixed `Main` entry point starts SysMonitor. No environment variable selects another system.
- Storage: SQLite /data/sysmonitor.db (SQLite with WAL on a data volume). Requests are handled on virtual threads with bounded parallelism.

2.2 Functional properties implementation

The following components implement the system:

Component	Responsibility
`SysMonitor.java`	Intake, subscriptions, history and management.
`MonitorStore.java`	SQLite store and outbox.
`Messaging.java`	AMQP and MQTT.

Every request passes `Service.handle`: request-count limit, `caller` (mutual TLS, certificate profile `sy` or `op`, identity token verified with Authentication and matched to the certificate), then the role's `route` with its caller lists.

2.2.1 Data handled and stored

Storage `SQLite /data/sysmonitor.db`:

Tables	Content
`observations, outbox`	History and output queue.
`subscriptions, amqp_bindings, audit`	Subscriptions, broker bindings and audit log.

The interfaces exchange JSON objects built at run time; their structure is specified in the IDD documents.

2.3 Non functional properties implementation

2.3.1 Security

`Tls` builds the TLS contexts from PEM files (`TLS_*`), with mandatory client certificates and CRL checking (`CRL_FILE`); `Service.caller` binds the identity token to the certificate; the Authentication client pins Authentication's certificate identity before sending credentials.

2.3.2 Power management

None; the system is a server process without power states. The `sysMonitor power` observation always reports line supply.

2.3.3 Internal monitoring

Logs go to stdout. The system publishes its `sysMonitor` observations every 30 s (see the sysMonitor IDD).

2.3.4 Configuration

Configuration is read from environment variables (listed in the SysD); secrets are read from files (`*_FILE` variables or mounted credential bundles).

2.4 Deployment

The local `Dockerfile` builds only this folder and runs `arrowhead-sysmonitor-standalone:5.2.2`. The local `compose.yaml` joins an operator-selected existing cloud network; it has no sibling build context, shared runtime image or dependency on another support process for startup. The HTTPS API uses port 8475; health uses internal port 8482. The process runs as UID 10001 with a read-only root filesystem and writable private `/data` and `/tmp` paths. The image preinstalls the platform SQLite native library in `/app/lib`; temporary storage remains non-executable. Certificates and credentials are mounted individually. This packaging is supplied for operator deployment; existing test-cloud deployments are separate deployment consumers and are not changed by it.

To select this system, copy `.env.example` to `.env`, provision its identity/password, system certificate/key, trusted CA and CRL, configure network and registry, then run `docker compose up -d --build` from this folder. The private project-scoped SQLite volume must never be shared with another system or concurrently mounted by a second instance. Stopping this Compose project preserves its volume and does not stop other systems.

ServiceRegistry is the bootstrap endpoint. Authentication and DynamicServiceOrchestration are looked up there; other consumed services are orchestrated through DSO and require the cloud's authorization rules. AMQPS monitoring remains a network dependency. The MQTT v5 output broker, observation intake and authorized orchestration must be available for operational readiness. SysMonitor reports degraded HTTP 503 when AMQP intake or MQTT output is disconnected. The other native systems return process liveness. Inspect registration, broker and orchestration status for operational readiness.

2.4.1 Ownership and build boundary

The helpers `Service`, `Tls`, `Json`, `Database` and `Messaging` are private source copies inside this project. The JAR contains no implementation of the other three former runtime systems. There is no shared runtime process, data directory or Maven parent. Maven dependencies remain external libraries. Security fixes to duplicated helpers must be propagated and verified in every affected project.

Build with Java 21 and `mvn -B clean verify`, or use the local Docker build. The packaged SQLite smoke check is `java -jar target/arrowhead-sysmonitor-5.2.2.jar storage-selftest`. The build requires no sibling support folder.

2.5 Local modifications

Notes on this implementation in the workspace:

- Native Java port (2026-10-16); bindings re-applied on every refresh (2026-10-19).

2.6 Verification

The project owns its JUnit security tests and system-specific tests. Verification covers mandatory mutual TLS, certificate/token binding, pinned Authentication identity, DSO discovery, concurrent token renewal and startup of this actual system with isolated generated credentials and its own temporary database. Federation and monitoring projects retain their relevant peer, session or persistence tests. The distribution verification report records independent build, packaging, image and configuration results; it does not claim a live-cloud rollout.

3 Services

The implementation of the services is based on the SD and IDD documents referenced below.

Services produced	**SysD ref**	**SD ref**	**IDD ref**
`sysMonitorManagement`	[1]	[2]	[3]
`sysMonitorHistory`	[1]	[4]	[5]
`sysMonitorData`	[1]	[6]	[7]
`sysMonitor`	[1]	[8]	[9]
Services consumed	**SysD ref**	**SD ref**	**IDD ref**
serviceDiscovery	[10]	[11]	–
identity	[12]	[13]	–
serviceOrchestration	[14]	[15]	–
authorization	[16]	[17]	–
sysMonitor (all providers)	AMQP broker	–	–
sysMonitorData	MQTT broker	–	–

4 References

[1] J. Delsing, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-SysMonitor-java-spring/documentation/v5.2.2/SysMonitor-v5.2.2-SysD.pdf.

[2] ——, “sysMonitorManagement – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorManagement-SD.pdf.

[3] ——, “sysMonitorManagement HTTP/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorManagement-HTTP-IDD.pdf.

[4] ——, “sysMonitorHistory – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorHistory-SD.pdf.

[5] ——, “sysMonitorHistory HTTP/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorHistory-HTTP-IDD.pdf.

[6] ——, “sysMonitorData – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorData-SD.pdf.

[7] ——, “sysMonitorData MQTT/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorData-MQTT-IDD.pdf.

[8] ——, “sysMonitor – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitor-SD.pdf.

[9] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitor-AMQP-IDD.pdf.

[10] ——, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-SysD.pdf.

[11] ——, “serviceDiscovery – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-serviceDiscovery-SD.pdf.

[12] ——, “Authentication – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-SysD.pdf.

[13] ——, “identity – Service Description (Authentication),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-identity-SD.pdf.

[14] ——, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-SysD.pdf.

[15] ——, “serviceOrchestration – Service Description (DynamicServiceOrchestration),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-serviceOrchestration-SD.pdf.

[16] ——, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.2.2/ConsumerAuthorization-v5.2.2-SysD.pdf.

[17] ——, “authorization – Service Description (ConsumerAuthorization),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.2.2/ConsumerAuthorization-v5.2.2-authorization-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Independent Java build, private helpers and state, per-system deployment and configuration.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	